# Does DVOL add information beyond HAR?

A retrospective BTC variance forecast study | 50kfunded | 19 October 2026

**Abstract.** I compare direct 30-day HAR forecasts with the same regression augmented by BTC DVOL, using 1,357 matched daily origins from 2023 to 2026. Augmentation lowers mean QLIKE by 11.35%, but its paired difference is -0.01424, with descriptive 95% interval [-0.03490, +0.00505] and estimated ESS 51.71, below the fixed threshold of 60. I find no clear improvement in a fixed variance-budget decision. My inference stress tests also show that one year of overlapping daily forecasts has no demonstrated useful confirmatory power under the declared gate, so I lock the future study as evidence collection and keep formal superiority claims disabled.

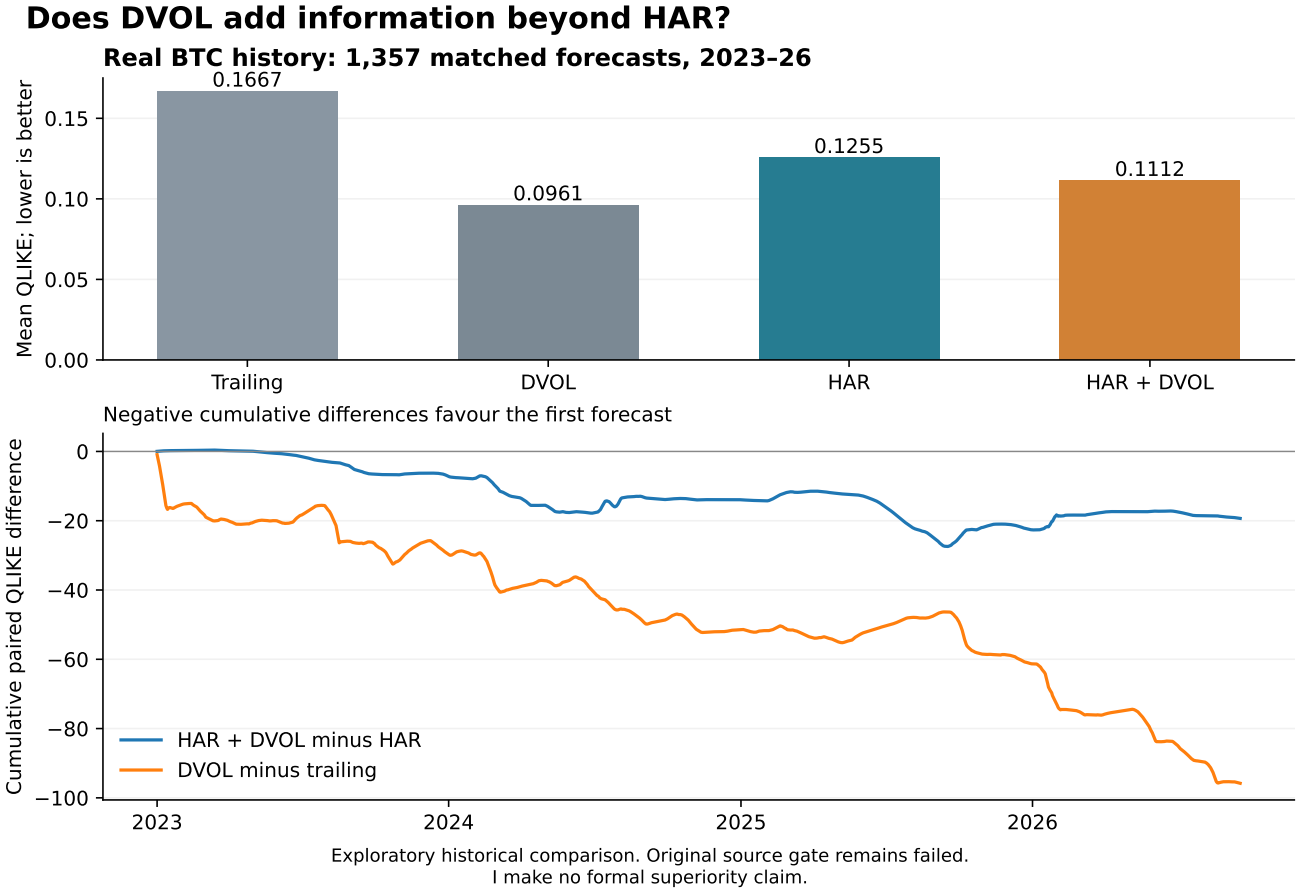



Figure 1. I use real retained historical inputs. Both panels use the same 1,357 origins. The lower panel shows error accumulation, rather than trading profit. All historical findings remain exploratory; the original vendor all-OHLC validation gate remains failed.

# 1. Question, inputs and matched design

I ask whether option-implied variance adds forecasting information beyond daily, weekly and monthly realised-variance components. I use Corsi's HAR construction as a parsimonious price-history benchmark, adapted to a direct 30-day target. I compare the benchmark with the same model plus previous completed daily DVOL variance. I also retain raw DVOL and trailing variance.

I downloaded 48,626 hourly BTC-PERPETUAL candles and 1,996 daily BTC DVOL candles on 19 October 2026. My matched evaluation uses daily midnight origins from 1 January 2023 through 18 September 2026. I use the next 720 hourly log-return squares, annualised by 8,760, as the target. Each 720-return window requires 721 completed closes. Historical downloads do not establish original unrevised vintages.

At each origin I fit both regressions using exactly the latest 365 labels whose full 30-day targets have ended. My HAR features are prior 1-, 7- and 30-day annualised variances; the augmented model adds squared DVOL/100. I use level OLS with an intercept. I centre and scale features using the training rows only. I fix a 1e-6 annual-variance floor, retain raw predictions, and count every floor hit. Neither model hits it in this sample.

$$
\begin{aligned}
V_t^{(30\mathrm{d})} &= \frac{8760}{720}\sum_{h=1}^{720} r_{t+h}^2, \\
RV_t^{(k)} &= \frac{8760}{24\,k}\sum_{h=1}^{24\,k} r_{t-h+1}^2, \quad k \in \{1,7,30\}, \\
\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR}} &= \widehat{\beta}_{0,t} + \widehat{\beta}_{d,t} RV_t^{(1)} + \widehat{\beta}_{w,t} RV_t^{(7)} + \widehat{\beta}_{m,t} RV_t^{(30)}, \\
\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR+DVOL}} &= \widehat{\gamma}_{0,t} + \widehat{\gamma}_{d,t} RV_t^{(1)} + \widehat{\gamma}_{w,t} RV_t^{(7)} + \widehat{\gamma}_{m,t} RV_t^{(30)} + \widehat{\gamma}_{D,t}\left(\frac{\mathrm{DVOL}_t^{\mathrm{prev.close}}}{100}\right)^2 .
\end{aligned}
$$

I score

$$
\mathrm{QLIKE} = \frac{\text{realised}}{\text{forecast}} - \log\left(\frac{\text{realised}}{\text{forecast}}\right) - 1.
$$

I compute every comparison on identical origins. The additional historical work follows inspection of these data, so chronological training does not make this an untouched experiment.

| Forecast | Mean QLIKE | Role |
|---|---|---|
| Trailing 30-day | 0.16670 | Reference |
| Raw DVOL | 0.09611 | Reference |
| HAR | 0.12548 | Benchmark |
| HAR + DVOL | 0.11123 | Augmentation |

I estimate an augmentation difference of -0.01424, with interval [-0.03490, +0.00505]. The interval includes zero and the ESS gate fails. Raw DVOL has the lowest point loss, which also exposes estimation risk in the fitted augmentation.

I include level Mincer-Zarnowitz regressions with HAC uncertainty, a joint HAR calibration/encompassing restriction, and a separate incremental-coefficient restriction with the HAR slope and intercept unrestricted. These generated-forecast diagnostics are exploratory; I do not treat their p-values as additional primary discoveries. Coefficients and uncertainty are in summary.json.

## 2. Loss concentration and a concrete decision

I find that the largest 1% and 5% of absolute augmentation loss differences account for 8.6% and 27.8% of total absolute differences. Their signed contributions are positive, so these largest observations hurt augmentation. The overall mean is still negative. This does not support a simple story that augmentation wins only by avoiding the most extreme losses.

| Year left out | Remaining origins | Mean augmentation difference |
|---|---|---|
| 2023 | 992 | -0.01233 |
| 2024 | 991 | -0.01260 |
| 2025 | 992 | -0.01072 |
| 2026 | 1096 | -0.02065 |

I retain contributions for every fixed calendar quarter, extreme-origin dates, and all 30 non-overlapping starting phases. The phase samples contain only 45 or 46 outcomes and are dependent. I use these displays to expose concentration, not to select a favourable crisis window or claim an ex-ante regime rule.

For decision relevance I fix an illustrative annual volatility budget of 20% and weight

$$w = \min\left(1, \frac{0.20}{\sqrt{\text{forecast variance}}}\right).$$

I measure the squared error

$$\left(w^2 \times \text{realised variance} - 0.04\right)^2.$$

This assumes constant fractional exposure for the variance calculation. I do not implement a fixed-unit portfolio, rebalancing, funding or profit, and this rule is not claimed optimal for squared budget error.

| Forecast | Mean weight | Mean squared budget error | Budget exceeded |
|---|---|---|---|
| Trailing | 0.469 | 0.0008186 | 46.7% |
| DVOL | 0.418 | 0.0002872 | 29.6% |
| HAR | 0.420 | 0.0004014 | 36.3% |
| HAR + DVOL | 0.437 | 0.0004196 | 41.7% |

I get augmented-minus-HAR risk loss +0.0000182, with interval [-0.0000818, +0.0001221]. Lower mean QLIKE does not establish a better decision under this loss. I report entry-plus-exit fee amounts separately at 0, 5, 10 and 20 basis points per side; I do not subtract costs from a risk error or infer trading returns.

## 3. Dependence and finite-sample inference

I use 5,000 paired stationary-bootstrap draws with a fixed seed, 60-day expected blocks and a 59-lag HAC check. I retain 30-, 90-, 180- and 365-day block sensitivities. The overlapping targets and rolling fits can create dependence beyond the 30-day target horizon. ESS is a diagnostic estimate, not an observed number of independent episodes.

I simulate 200 independent samples per declared scenario at n=365 and n=1,357, using 2,000 bootstrap draws each. My population null is exactly mean zero without centring each realised sample. I use a 30-day moving-average loss component, a stationary slow component with persistence exp(-1/365), and symmetric rare shocks spread across overlapping windows. The synthetic alternative is a -0.25 marginal-standard-deviation shift, not -0.25 QLIKE. These stress tests approximate loss dependence and do not simulate the entire rolling HAR estimation pipeline.

| n / synthetic process | Ungated rejection | 95% MC interval | ESS gate passes |
|---|---|---|---|
| 365 / Overlap | 19.0% | 14.2% to 25.0% | 0.0% |
| 365 / Overlap + slow | 49.0% | 42.2% to 55.9% | 0.0% |
| 365 / Overlap + tails | 16.0% | 11.6% to 21.7% | 0.0% |
| 1357 / Overlap | 10.0% | 6.6% to 14.9% | 1.5% |
| 1357 / Overlap + slow | 45.0% | 38.3% to 51.9% | 0.0% |
| 1357 / Overlap + tails | 8.5% | 5.4% to 13.2% | 1.5% |

At n=365 I observe ungated null rejection rates of 16% to 49%, rather than nominal 5%. The ESS gate passes zero of 200 replications under every one-year null and alternative, giving a Wilson 95% upper bound around 1.9% for the pass probability. I therefore have no demonstrated useful confirmatory power for the planned year. At n=1,357, conditional estimates sometimes use only one to four passing replications; I retain their wide Monte Carlo intervals.

I committed a separate inference-validation amendment that disables formal superiority claims for the new HAR period. I preserve its fixed terminal date and primary contrast, collect timestamped evidence, and accept an inconclusive terminal result. A later inference design needs its own prospective declaration; I will not change this cohort to recover significance.

## 4. Prospective collection, limits and reproduction

I locked 365 future HAR origins from 21 October 2026 through 20 October 2027, each at 01:00 UTC. Features end at midnight; I freeze between 00:10 and 00:50, then start the outcome at 01:00. I train on the correspondingly shifted historical 30-day targets completed by midnight. This differs from the midnight target alignment in the retrospective comparison. Final maturity is 19 November 2027; I fix the terminal analysis at 21 November 2027, 01:00 UTC.

I require a matching hourly and daily DVOL close for each new forecast, preserve raw capture segments, attach code and input hashes, and check fitting and durable-write completion against the freeze deadline. I count missed scheduled days and do not backfill. I withhold terminal intervals unless all 365 planned pairs and source checks exist; late-arriving scores after the fixed analysis cutoff stay excluded.

My separate SVI study still has zero settled research forecasts. Its original protocol and baseline locks remain unchanged. A versioned amendment fixes its expiry cutoff at 20 October 2027, 08:00 UTC and final decision at 22 October 2027, 08:00 UTC. Both reporting and CLI inference keep earlier output descriptive and pin terminal data availability. I retain calendar-grid and extrapolated-tail limitations. Local forecast hash chains are not independent public timestamps.

My collector retries recoverable delivery pagination in fresh segments and keeps conflicting immutable prices fatal. I cache cleaning by input, protocol, code and runtime fingerprints, and render reports outside the forecast loop. I publish counts, rejection reasons, local hashes and heartbeat age. A fresh heartbeat does not prove process health. Continuous collection needs an awake online host; short validation recordings do not establish sustained operation.

The original all-OHLC audit remains failed because one unused DVOL open differs. All three audited DVOL closes match, but these dates do not validate the entire history. I use perpetual returns as an index proxy, a single asset and venue, retrieved vintages and overlapping outcomes. I do not identify a causal variance premium or executable edge.

**Reproduction.** I use Python 3.12 with requirements-research.txt, then run **python scripts/build_research.py**. I retain the full report, matched forecasts, coefficients, training cutoffs, diagnostics, simulation rates and plan hashes. The command reads the checked-in public recordings and makes no market requests.

## References.


Corsi (2009), A Simple Approximate Long-Memory Model of Realized Volatility, doi:10.1093/jjfinec/nbp001.

Patton (2011), Volatility Forecast Comparison Using Imperfect Volatility Proxies, doi:10.1016/j.jeconom.2010.03.034.

Politis and Romano (1994), The Stationary Bootstrap, doi:10.1080/01621459.1994.10476870.

Deribit, DVOL methodology, official methodology.